Dynamics: Problem Sheet 2 (of 8)

1. Consider a particle of mass m moving vertically in a fluid with quadratic drag force Dv^2, where v is its velocity and $D > 0$ is a constant. The particle is also acted on by gravity, with acceleration due to gravity g.

   (a) Consider dropping the particle from rest through the fluid, so that its velocity is $v = \dot{z} \leq 0$, with z measured upwards. Show that the equation of motion may be written as

   $$m\dot{v} \;=\; -mg + Dv^2 \;.$$

   Show that this may be integrated to

   $$t \;=\; -\int_0^v \frac{\mathrm{d}u}{g - \frac{Du^2}{m}} \;.$$

   By evaluating the integral, hence show that the solution is

   $$v(t) \;=\; -\sqrt{\frac{mg}{D}}\,\tanh\left(\sqrt{\frac{Dg}{m}}\,t\right) \;.$$

   What is the terminal velocity?

   (b) Now consider projecting the particle upwards through the fluid, starting at $z = 0$ with speed u. Show that the equation of motion may be written as

   $$\frac{\mathrm{d}(v^2)}{\mathrm{d}z} \;=\; 2\dot{v} \;=\; -2g - \frac{2Dv^2}{m} \;.$$

   Regarding this as an equation for $v^2(z)$, by integrating it show that the maximum height reached is

   $$z_{\max} \;=\; \frac{m}{2D}\log\left(1 + \frac{Du^2}{mg}\right) \;.$$

   What happens as $D \to 0$?

2. The Rotor is an amusement park ride shaped like a large cylindrical drum. Riders stand on the inside, with their back against the cylinder, which then begins to rotate. Once at full speed, the floor drops out, leaving riders stuck to the rotating wall.

   We consider a rider as a point mass of mass m at position $\mathbf{r}(t) = (R\cos\theta(t), R\sin\theta(t), z(t))$, where R is the radius of the cylinder. When attached to the wall without slipping, we define $\dot{\theta}(t) = \Omega$ as the rotation rate at full speed. Newton's 2nd law reads $m\ddot{\mathbf{r}} = \mathbf{N} - mg\mathbf{e}_z + \mathbf{F}$, where $\mathbf{N}$ is a normal reaction force, pointing in the negative of the radial direction $\mathbf{e}_r = (\cos\theta, \sin\theta, 0)$, and $\mathbf{F}$ is a friction force.

   (a) Show that if there is *no* friction force, then riders will not remain stuck to the wall when the floor is retracted, but rather will drop.

   (b) In reality, the wall provides a friction force $\mathbf{F} = F\mathbf{e}_z$, opposing gravity. Under so-called dry friction, the body will not slip so long as

   $$F \leq \mu N,$$

where $N = |\mathbf{N}|$ and μ is a friction coefficient.
Show that this condition translates to

$$\Omega \geq \sqrt{\frac{g}{\mu R}}.$$

Why is a person's mass irrelevant to whether or not they will slip? If $R = 3$ m, and $\mu = 1$, how fast must the drum rotate?

3. A particle of mass m moves along the x axis with one end attached to a spring of spring constant $k > 0$, and is subjected to an additional force $F_0 \cos \Omega t$.

   (a) Show that the equation of motion is

   $$\ddot{x} + \omega^2 x \;=\; A \cos \Omega t \;,$$

   where $x = 0$ corresponds to the unstretched position of the spring, $\omega = \sqrt{k/m}$, and $A = F_0/m$.

   (b) Suppose that $x = \dot{x} = 0$ at time $t = 0$. Derive the solution

   $$x(t) \;=\; \frac{A}{\omega^2 - \Omega^2} \left(\cos \Omega t - \cos \omega t\right)$$

   for $\Omega \neq \omega$ and the solution

   $$x(t) \;=\; \frac{A}{2\omega} t \, \sin \omega t$$

   for $\Omega = \omega$. What is the qualitative difference between the two solutions?

4. Consider a particle of charge q moving in a constant electromagnetic field. Without loss of generality we take the magnetic field $\mathbf{B} = (0, 0, B) \neq 0$ to point along the z axis, while the electric field $\mathbf{E} = (E_1, E_2, E_3)$ is constant, but arbitrary.

   (a) Assuming the particle has mass m, but ignoring gravity, show that Newton's second law implies the coupled ODEs

   $$\begin{aligned} m\ddot{x} &= q\, E_1 + qB\, \dot{y} \;, \\ m\ddot{y} &= q\, E_2 - qB\, \dot{x} \;, \\ m\ddot{z} &= q\, E_3 \;, \end{aligned}$$

   for the position $\mathbf{r} = (x, y, z)$ of the particle.

   (b) Verify that

   $$\begin{aligned} x(t) &= x_0 + \frac{E_2}{B}\, t + R \cos(\omega t + \phi) \;, \\ y(t) &= y_0 - \frac{E_1}{B}\, t - R \sin(\omega t + \phi) \;, \\ z(t) &= z_0 + u\, t + \frac{q}{2m} E_3\, t^2 \;, \end{aligned}$$

   solves the equations of motion in part (a), where $\omega = qB/m$ is the cyclotron frequency, (x_0, y_0, z_0) is a constant vector, and u, R and ϕ are also constants.
   [*Optional*: For a more challenging version of this question, rather than verifying the solution, instead *derive* it, hence showing it is the general solution.]

   (c) Describe the path of the particle.